Hung Le

Doctoral Candidate at Aalto University, Finland



Personal Information

- Email: hung.h.le@aalto.fi
- Website: https://hungleeconomics.github.io/

Education

- **Aalto University** — Espoo, Finland
  Doctor of Science in Economics and Business Administration, Economics; GPA: 4.59/5 — *Sep 2022 - 2027 (Expected)*
  - **Chapter 1**: Competition or Collusion? Entry Decisions in the Swedish Pharmaceutical Market (with Otto Toivanen) CEPR Discussion Paper DP21676, VoxEU Column
  - **Chapter 2**: Reference Pricing as a Focal Point: Evidence from the Finnish Pharmaceutical Market (with Matias Pousi)
  - **Chapter 3**: A Stylized Logit Demand Model with Internal Reference Pricing
- **Aalto University** — Espoo, Finland
  Master of Science, Economics, Research Track; GPA: 4.58/5 — *Sep 2019 - Apr 2022*
- **Aalto University** — Mikkeli, Finland
  Bachelor of Science, International Business; GPA: 4.93/5 — *Sep 2016 - Feb 2019*

Experience

- **Aalto University** — Espoo, Finland
  Teaching Assistant
  - **Math Camp, Ph.D. level**: Aug 2024, Aug 2025
  - **Empirical Industrial Organization, M.Sc. level**: Oct 2024, Oct 2025
  - **Intermediate Macroeconomics II, B.Sc. level**: Mar 2023, Mar 2024, Mar 2025
  - **Principles of Economics, B.Sc. level**: Sep 2022, Sep 2023, Sep 2024, Sep 2025
  - **Mathematics for Economists, B.Sc. level**: Oct 2021
- **Aalto University** — Espoo, Finland
  Research Assistant — *Jun 2022 - August 2022*
- **VATT Institute for Economic Research (Valtion taloudellinen tutkimuskeskus)** — Helsinki, Finland
  Summer Trainee / Research Assistant — *May 2021 - May 2022*

Job Market Paper

- **Reference Pricing as a Focal Point: Evidence from the Finnish Pharmaceutical Market** with Matias Pousi

Abstract:

While price regulation is widely used to decrease prices, it can inadvertently provide firms with a focal point for coordination. The reference price system adopted in the Finnish pharmaceutical market in 2009 initially resulted in lower prices, but after a few years, prices gradually returned to their pre-reform level. How did firms succeed in raising prices, and what role did the regulation play in this process? We find that firms learned to use the previous quarter's reference price as a focal point when choosing prices at the beginning of each quarter. This pattern led to price increases from quarter to quarter. Six big generic firms were more likely to use focal-point pricing, and their presence was associated with similar behavior among smaller rivals. According to our structural model, 65% of products had a profitable downward deviation from the upward pricing path but only 13% deviated. The remaining 52% suggests potential tacit coordination. On the positive side, we show that after the quarterly reference price was fixed at the beginning of the quarter, it constrained firms from increasing prices higher.

References

- Professor Otto Toivanen, Aalto University (main supervisor)
  otto.toivanen@aalto.fi
- Professor Pauli Murto, Aalto University
  pauli.murto@aalto.fi
- Professor Alina Ozhegova, Aalto University
  alina.ozhegova@aalto.fi

Honors and Awards

- Dean's List, Dec 2019
- Ilkka Kontula Foundation Awards, May 2018

Grants Awarded

- OP Research Foundation, Mar 2025, Mar 2026
- Yrjö Jahnsson Foundation, Dec 2023
- Jenny and Antti Wihuri Foundation, Sep 2023
- Foundation for Economic Education Liikesivistysrahasto, May 2022, Nov 2024, Nov 2025

Skills

- Software: Stata, LaTeX, Microsoft 365, Python, R
- Language: English (fluent), Vietnamese (native), Finnish (basic)

Conferences and Seminars

- Reference Pricing as a Focal Point: Evidence from the Finnish Pharmaceutical Market (Job Market Paper)

  European Commission DG-Comp (Brussels, 2026, invited), Finnish Competition and Consumer Authority (online, 2026, invited), NORIO (Copenhagen, 2026), European Winter Meeting of the Econometric Society (Warsaw, 2026, upcoming)

- Competition or Collusion? Entry Decisions in the Swedish Pharmaceutical Market

  European Commission DG-Comp (Brussels, 2026, invited), CEPR Health Economics (Toulouse, 2026), EARIE (Mannheim, 2026), CRESSE (Kos, 2026), Swedish Competition and Consumer Authority (online, 2026, invited, upcoming)

Working Papers

- **Competition or Collusion? Entry Decisions in the Swedish Pharmaceutical Market** with Otto Toivanen

Abstract:

Price regulation may have unintended consequences, such as facilitating collusion or softening actual price competition by causing a reduction in the number of active firms. Observing different firms having a monopoly position in parts of a market may be the result of either intense post-entry competition and non-cooperative entry decisions, or collusive entry. This phenomenon and large price differences per pill within a market are widespread in the Swedish generic pharmaceutical market where price regulation through monthly auctions for each active ingredient – dosage form – strength – package size combination channels a large part of demand to the winner within each such substitution group. 22% of the markets are potentially collusive in having at least two firms as monopolies for different substitution groups at least some of the time and prices per pill in such markets are on average significantly higher. We take a structural model tailored to the Swedish circumstances to data on two markets. In one market, the entry patterns (monopoly package sizes) are suggestive of actual, in the other of attempted, collusion. Our counterfactual analysis where we induce substitution across package sizes yield savings of ¿50% in the first market, but lead to a modest expenditure increase in the other because intensified post-entry competition induces exit.

CEPR Discussion Paper DP21676, VoxEU Column

- **A Stylized Logit Demand Model with Internal Reference Pricing**

Abstract:

I study conditions for the existence and uniqueness of equilibrium in a logit model that incorporates an internal reference price and a copayment structure. Using the Intermediate Value Theorem, I prove equilibrium existence in a two-product market and show that the equilibrium is unique under additional restrictions: the two firms must be sufficiently different so that their best-response functions cross but do not overlap over an interval. I then extend the existence result to an n-product market with an aggregative-game approach.

Work in Progress

- **Entry Decisions in the Swedish Pharmaceutical Market** with Otto Toivanen

  Abstract:

  We extend our analysis of entry decisions in the Swedish pharmaceutical market to include more substitution groups.

- **The flow of raw materials in Finland**

  Abstract:

  I study how prices affect the flow of raw materials from production plants to manufacturing plants in Finland.